Smoothed projections in FEM reading group

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Disclaimer. For pedagogical reasons, the following note contains many simplifications that may not be entirely realistic. However, all of them can be overcome to obtain a useful construction.

1 Motivation

a. Often, the natural interpolator[1] deduced from the degrees of freedom is not well defined in the expected space. Said differently, they can only be applied to fields enjoying greater regularity.

Example 1.1 (Lagrange). *The space $\mathcal{L}_1(\mathcal{T}) \subset \mathrm{H}^1(\Omega)$ has canonical interpolator*

$$Q_{1,\mathcal{T}}(u) = \sum_{x \in \mathcal{T}^0} u(x)\lambda_x,$$

where $\mathcal{T}^k$ are the simplices of dimension k and λ_x is the barycentric coordinate of x, i.e., $\lambda_x(x') = \delta_{x,x'}$. $Q_{1,\mathcal{T}}$ is only well-defined if $u|_K(x) = u|_{K'}(x)$ whenever K and K' share the vertex x. This holds when u is continuous. Nevertheless, $u \in \mathrm{H}^1(\Omega)$ does not necessarily admit a continuous representative. Indeed, the Sobolev embedding theorem says that $H^s(\Omega) \hookrightarrow C^0(\Omega)$ when $s > n/2$. Therefore $Q_{1,\mathcal{T}}$ is defined in all of $\mathrm{H}^1(\Omega)$ only when $n = 1$. But there the finite element method usually coincides with other methods. In general, we only have

$$Q_{1,\mathcal{T}} : \mathrm{H}^{n/2+\epsilon}(\Omega) \to \mathcal{L}_1(\mathcal{T}).$$

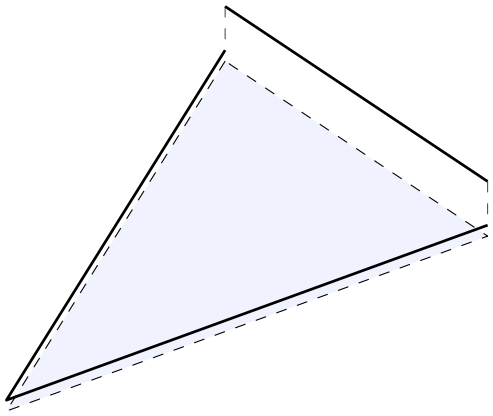
Example 1.2 (Raviart-Thomas). *The space $\mathbf{RT}_0(\mathcal{T}) \subset \mathbf{H}(\mathsf{div}; \Omega)$ has canonical interpolator*

$$\Pi_{0,\mathcal{T}}(\boldsymbol{\tau}) = \sum_{f \in \mathcal{T}^{n-1}} \left(\int_f \boldsymbol{\tau} \cdot \mathbf{n} \right) \boldsymbol{\tau}_f \,,$$

where $\left\{ \boldsymbol{\tau}_f \right\}_f$ is the basis associated with the moments $\int_f \boldsymbol{\tau} \cdot \mathbf{n}$. For this to make sense, we need the normal trace $\boldsymbol{\tau} \cdot \mathbf{n}$ to be "localisable" and integrable.

An intuitive approach to get rid of the second condition would be to define the moments using the duality pairing $\langle \boldsymbol{\tau} \cdot \mathbf{n}, 1 \rangle_f$. Unfortunately, as a functional in $\mathrm{H}^{-1/2}(\partial K)$, the restriction of $\boldsymbol{\tau} \cdot \mathbf{n}$ to f only makes sense against functions in $\mathrm{H}^{1/2}_{00}(f)$. That is, functions whose extension by zero to ∂K belongs to $\mathrm{H}^{1/2}(\partial K)$, or equivalently, admits an H^1-lifting to all of K. The function $f \ni x \mapsto 1$ clearly doesn't satisfy this, as it's observed in the following picture:

[1] Usually called interpolation operators, but for the sake of brevity we follow Snorre's naming convention.



This says that we cannot dispense with integrability in order to have moments supported on faces.

A second approach: define the moments on regular functions and prove they are bounded, then they must admit a unique extension to the completion of the space. With lots of effort, the best local bound we can have is:

Lemma 3.28. *Let $t \in (1, +\infty)$ and $\delta \in [0, 1]$ such that*

$$\begin{cases} \delta > \dfrac{1}{t} & \text{if } t \in (1, 2]\,, \\ \delta \geqslant 0 & \text{if } t \in (2, +\infty)\,. \end{cases}$$

Then, there exists a constant $\widehat{C} > 0$, independent of h, such that

$$\|\Pi^k_{\widehat{K}}(\widehat{\boldsymbol{\tau}})\|_{0,t;\widehat{K}} \leqslant \widehat{C} \left\{ \|\widehat{\boldsymbol{\tau}}\|_{\delta,t;\widehat{K}} + \|\mathrm{div}(\widehat{\boldsymbol{\tau}})\|_{0,t;\widehat{K}} \right\} \qquad \forall\, \widehat{\boldsymbol{\tau}} \in \mathbf{W}^{\delta,t}(\widehat{K}) \cap \mathbf{H}^t(\mathrm{div}_t; \widehat{K})\,.$$

In our case, $t = 2$, so we are forced to take $\delta > 1/2$, obtaining

$$\Pi_{\hat{K}} : \mathbf{H}^{\frac{1}{2}+\epsilon}(\hat{K}) \cap \mathbf{H}(\mathrm{div}; \hat{K}) \to \mathbf{RT}_0(\hat{K})\,, \qquad \|\Pi_{\hat{K}} \hat{\boldsymbol{\tau}}\|_{0,\hat{K}} \leq \hat{C} \left\{ \|\hat{\boldsymbol{\tau}}\|_{\mathrm{div},\hat{K}} + |\hat{\boldsymbol{\tau}}|_{\frac{1}{2}+\epsilon,\hat{K}} \right\}\,,$$

a small perturbation of the $\mathbf{H}(\mathrm{div})$ norm.

The space $\mathbf{H}^{\frac{1}{2}+\epsilon}(\hat{K}) \cap \mathbf{H}(\mathrm{div}; \hat{K})$ is already complete in the norm in the RHS, therefore **we cannot extend $\Pi_{\hat{K}}$ to all of $\mathbf{H}(\mathrm{div}; \hat{K})$** *by a density argument.*

The estimate in the previous example will also be relevant in the following point:

b. The Raviart-Thomas interpolator does not enjoy **enough continuity properties**. That is, it is not bounded in the expected norm. Despite the obvious reason that it would be expected because $\mathbf{RT}_0(\mathcal{T})$ is the canonical[2] way of discretising $\mathbf{H}(\mathrm{div}; \Omega)$, there's a deeper reason related with stability of the scheme: We know from FEEC that the well-posedness of a Hodge-Laplace problem amounts to the existence of Poincaré inequalities. Thus, stability of a Galerkin scheme reduces to finding such inequalities uniform in h.

Example 1.3. *The Dirichlet Poisson problem corresponds to the last Hodge laplacian problem:*

$$\begin{cases} \mathrm{div}(-\nabla u) = \mathrm{f} & \text{in } \Omega \\ u = 0 & \text{on } \partial\Omega \end{cases} \iff \begin{cases} (\boldsymbol{\sigma}, \boldsymbol{\tau}) - (u, \mathrm{div}\, \boldsymbol{\tau}) = 0 & \forall\, \boldsymbol{\tau} \in \mathbf{H}(\mathrm{div}; \Omega) \\ (\mathrm{div}\, \boldsymbol{\sigma}, v) = (\mathrm{f}, v) & \forall\, v \in \mathrm{L}^2(\Omega) \end{cases}$$

[2] Don't believe **me**, believe Kaibo and Ting [5]: *"On the discrete level, the Whitney forms for the de Rham complex and the dual Whitney forms turn out to be the canonical discretization of the de Rham complexes."*, or Ralf [4]: *"In this paper we have developed a fully canonical construction of simplicial finite elements[...]"*

This is well posed because of the Poincaré inequality $\|\boldsymbol{\tau}\|_{\mathsf{div};\Omega} \leq C_P \|\,\mathsf{div}\,\boldsymbol{\tau}\|_{0,\Omega}$*, for all* $\boldsymbol{\tau} \perp \ker(\mathsf{div})$.

Suppose for a moment we have

$$\Pi_{0,\mathcal{T}} : \mathbf{H}(\mathsf{div};\Omega) \to \mathbf{RT}_0(\mathcal{T})\,, \qquad \|\Pi_{0,\mathcal{T}}\boldsymbol{\tau}\|_{\mathsf{div},\Omega} \leq C_\Pi \|\boldsymbol{\tau}\|_{\mathsf{div},\Omega}\,,$$

for C independent of h, and a **commuting property** $\mathsf{div}\,\Pi_{0,\mathcal{T}} = \mathrm{P}_{0,\mathcal{T}}\,\mathsf{div}$*, where the latter is the projector over* $\mathsf{DG}_0(\mathcal{T})$*. Then, the uniform Poincaré inequality holds. Explicitly*

$$\|\boldsymbol{\tau}_h\|_{\mathsf{div};\Omega} \leq C_\Pi\, C_P \|\,\mathsf{div}\,\boldsymbol{\tau}_h\| \qquad \forall\, \boldsymbol{\tau}_h \perp \ker(\mathsf{div})$$

Therefore, **the Galerkin approximation is well-posed and stable**, *which implies that the Galerkin projection is uniformly bounded.*

Remark 1.1. *In general, for any Hilbert complex one has*

> **Theorem 5.3 (discrete Poincaré inequality).** *Assume that the complex* (5.5) *is a finite dimensional subcomplex of* (5.1) *which admits a V-bounded cochain projection π_h. Then*
>
> $$\|v\|_V \leq c_P \|\pi_h^k\|\, \|dv\|_V, \quad v \in \mathfrak{Z}_h^{k\perp} \cap V_h^k,$$
>
> *where c_P is the constant appearing in the Poincaré inequality* (4.7) *and $\|\pi_h^k\|$ denotes the V^k operator norm of π_h^k.*

The most natural solution to the fact that interpolators only can be evaluated in regular fields is to **first smooth** and **then interpolate**, so that they are always well-defined. The tricky part is to show that this naive approach does not break any of the desired properties.

To summarise, the *desired properties* are:

1. $\Pi_h \boldsymbol{\tau} = \boldsymbol{\tau}$ for all $\boldsymbol{\tau} \in \mathbf{RT}_0(\mathcal{T})$
2. $\mathsf{div}\,\Pi_h = \mathrm{P}_h\,\mathsf{div}$, where P_h is a projection onto $\mathsf{DG}_0(\mathcal{T})$
3. $\Pi_h : \mathbf{H}(\mathsf{div};\Omega) \to \mathbf{RT}_0(\mathcal{T})$ is well-defined and bounded
4. Π_h satisfies optimal approximation properties. For $\mathbf{RT}_0(\mathcal{T})$, this means

$$\forall\, \boldsymbol{\tau} \in \mathbf{H}^1(\Omega)\ :\ \|\Pi_h\boldsymbol{\tau} - \boldsymbol{\tau}\|_{0,\Omega} \preceq h\|\nabla\boldsymbol{\tau}\|_{0,\Omega}\,.$$

2 Construction of smoothed projections

2.1 Smoothing

Choose a mollifier ρ supported in $\mathbb{B}_1(0)$ and use the scaling $\rho_\epsilon = \epsilon^{-n}\rho(\cdot/\epsilon)$ to define

$$\mathcal{R}^\epsilon\boldsymbol{\tau}[x] = \int_{\mathbf{R}^n} \rho_\epsilon(x-y)\boldsymbol{\tau}(y)\,dy = \int_{\mathbf{R}^n} \rho_\epsilon(y)\boldsymbol{\tau}(x-y)\,dy\,.$$

One immediately notices that $\rho_\epsilon(x-y)$ vanishes whenever $y \notin \mathbb{B}_\epsilon(x)$. This implies that $\mathcal{R}^\epsilon$ is quasi-local: **The value $\mathcal{R}^\epsilon\boldsymbol{\tau}[x]$ only depends on the values of $\boldsymbol{\tau}$ on a neighbourhood of x.**

More precisely, if $\mathcal{V}^\epsilon(U)$ denotes the ϵ-neighbourhood of a set $U \subset \mathbf{R}^n$, and $\boldsymbol{\mathcal{F}}(U)$ the set of all vector fields $U \to \mathbf{R}^n$. Then, $\mathcal{R}^\epsilon$ maps $\boldsymbol{\mathcal{F}}(\mathcal{V}^\epsilon(U))$ into $\boldsymbol{\mathcal{F}}(U)$, whenever well defined.

It is straightforward to show that

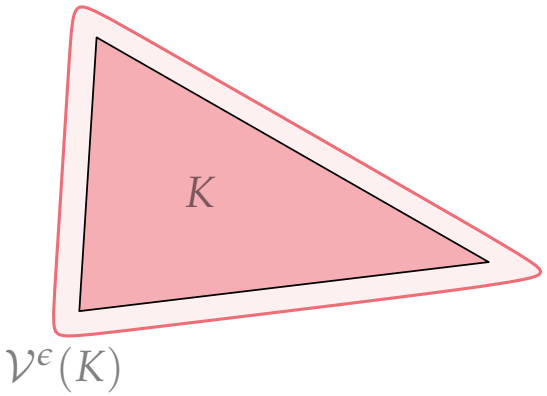
$$\mathcal{R}^\epsilon : \mathbf{L}^1_{\mathrm{loc}}(\mathcal{V}^\epsilon(U)) \to \mathbf{C}^\infty(U) \quad \text{and} \quad \operatorname{div}\mathcal{R}^\epsilon\boldsymbol{\tau} = \mathcal{R}^\epsilon \operatorname{div}\boldsymbol{\tau}\,.$$

When h is the maximum diameter in $\mathcal{T}$, we write $\mathcal{R}^\epsilon_h = \mathcal{R}^{\epsilon h}$. Throughout, Ω is a bounded polytope and $\mathcal{T}$ is quasi-uniform, i.e. $h_K \simeq h$ for all $K \in \mathcal{T}^n$. In particular, $\mathcal{V}^\epsilon_h(K) \subset \mathcal{M}(K)$ for ϵ small.

Remark 2.1. *Working with differential forms is a more natural approach, in which we can do a unified construction for the entire de Rham complex. In that case, if $\mathrm{t}_y(x) = x - y$ is the translation, one defines*

$$\mathcal{R}^\epsilon\omega = \int_{\mathbf{R}^n} \phi_\epsilon\, \mathrm{t}_y^*\omega\,.$$

According to the non-local nature of $\mathcal{R}^\epsilon$, we have to take extra care of the cells touching the boundary. This is because $\mathcal{V}^\epsilon(K) \not\subset \Omega$ for every cell $K \in \mathcal{T}^n$ intersecting the boundary. This implies that we cannot smooth within these cells. This motivates introducing an extension of our fields.



2.2 Extension

We need to distinguish those cells intersecting the boundary. Thus, we collect them in

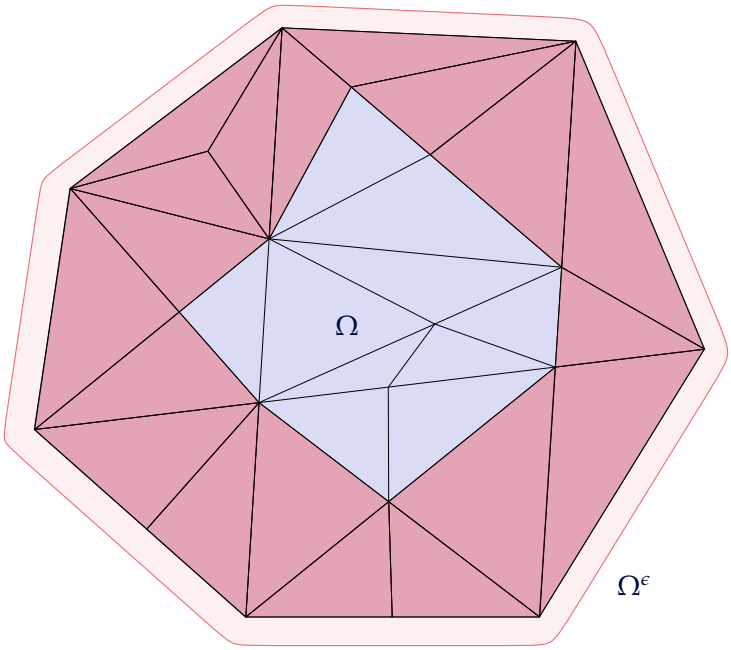
$$\mathcal{T}^n_\partial = \{K \in \mathcal{T}^n \mid K \cap \partial\Omega \neq \emptyset\}\,.$$

This induces a proper simplicial complex $\mathcal{T}_\partial = \left\{K' \in \mathcal{T} \,:\, \exists K \in \mathcal{T}^n_\partial \,:\, K' \lhd K\right\}$, which justifies the notation $\mathcal{T}^n_\partial$. Define the "interior complex" $\mathcal{T}_i$ to be the one induced by $\mathcal{T}^n_i = \mathcal{T}^n \backslash \mathcal{T}^n_\partial$.

Letting $\mathcal{M}(K)$ denote the macroelement of K and putting $\Omega^\epsilon = \mathcal{V}^\epsilon(\Omega)$, we assume we have an extension operator

$$\mathcal{E}^\epsilon : \boldsymbol{\mathcal{F}}(\Omega) \to \boldsymbol{\mathcal{F}}(\Omega^\epsilon)\,, \qquad \mathcal{E}^\epsilon\boldsymbol{\tau} = \boldsymbol{\tau} \quad \text{on } \Omega\,.$$

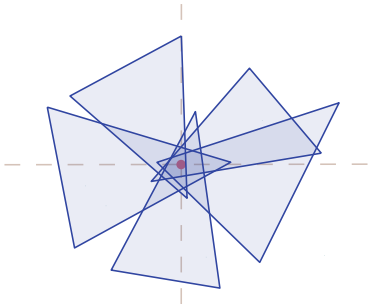
As for the regularisers, we write $\mathcal{E}_h^\epsilon = \mathcal{E}^{\epsilon h}$ and $\Omega_h^\epsilon = \Omega^{\epsilon h}$. We require $\mathcal{E}^\epsilon$ to be local, in that $(\mathcal{E}^\epsilon \boldsymbol{\tau})|_{\mathcal{V}^\epsilon(K)}$ depends only on $\boldsymbol{\tau}|_{\mathcal{V}^\epsilon(K)\cap\Omega}$. Thus $\mathcal{E}^\epsilon$ only "creates" values in the tubular neighbourhood $\Omega^\epsilon \backslash \Omega$.



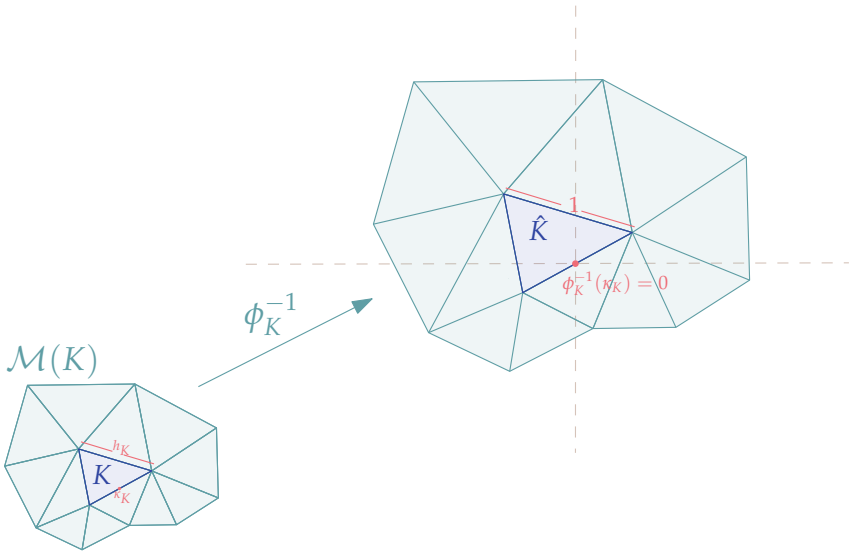
Since $\mathcal{E}^\epsilon \boldsymbol{\tau} = \boldsymbol{\tau}$ on Ω, no extension is actually used on cells $K \in \mathcal{T}_i^n$ with $\mathcal{V}^\epsilon(K) \subset \Omega$, and for all $K \in \mathcal{T}^n$ the composition $\mathcal{R}_h^\epsilon \mathcal{E}_h^\epsilon : \boldsymbol{\mathcal{F}}(\Omega) \to \mathbf{C}^\infty(K)$ is well-defined.

2.3 Scaling

In what follows we use **scaling by dilation and translation**[3]. Instead of mapping all the top-dimensional cells to the unit simplex, we obtain a family of reference cells in a compact neighborhood of the origin. Formally, for any cell K, we choose an interior point $\kappa_K \in K$ uniformly far from ∂K (e.g. the barycentre), and define $\phi_K x = hx + \kappa_K$, and set $\hat{K} = \phi_K^{-1}(K)$ and $\hat{\boldsymbol{\tau}} = \phi_K^* \boldsymbol{\tau}$.



[3]This is how Arnold, Falk and Winther called it in their foundational paper. Though for us it's more "contraction and translation" because the maps are $\hat{K} \to K$



I learned from Snorre that a major advantage of this approach is that the smoothing is stable under pullback[4] by ϕ_K, which is to say that smoothing commutes with pullback:

$$\begin{array}{ccc} \mathcal{V}^\epsilon(\hat{K}) & \xrightarrow{\phi} & \mathcal{V}_h^\epsilon(K) \end{array}$$

$$\begin{array}{ccc} \boldsymbol{\mathcal{F}}(\mathcal{V}^\epsilon(\hat{K})) & \xleftarrow[\phi^*]{} & \boldsymbol{\mathcal{F}}(\mathcal{V}_h^\epsilon(K)) \\ \downarrow{\scriptstyle \mathcal{R}^\epsilon} & & \downarrow{\scriptstyle \mathcal{R}_h^\epsilon} \\ \mathbf{C}^\infty(\hat{K}) & \xleftarrow[\phi^*]{} & \mathbf{C}^\infty(K) \end{array}$$

$$\begin{array}{ccc} \hat{K} & \xrightarrow{\phi} & K \end{array}$$

By a change of variables and the chain rule one finds out that

$$\|\hat{\boldsymbol{\tau}}\|_{0,\hat{K}} = h^{(n-1)-k/2}\|\boldsymbol{\tau}\|_{0,K} \quad \text{and} \quad \|\nabla\hat{\boldsymbol{\tau}}\|_{0,\hat{K}} = h^{n-k/2}\|\nabla\boldsymbol{\tau}\|_{0,K}\,,$$

for any $K \in \mathcal{T}^k$. Both estimates are referred to as **scaling**. The typical way of using this is proving an estimate of the form

$$\|\mathcal{S}_{\hat{K}}\hat{\boldsymbol{\tau}}\|_{0,\hat{K}} \leq C(\hat{K})\|\nabla\hat{\boldsymbol{\tau}}\|_{0,\hat{K}}$$

in a reference element $\hat{K}$, and then scaling back to get

$$\|\mathcal{S}_K\boldsymbol{\tau}\|_{0,K} \leq hC(\hat{K})\|\nabla\boldsymbol{\tau}\|_{0,K}\,,$$

where one gains a power of h because of the difference in the scaling identities.

One needs to ensure that the constant $C(\hat{K})$ can be chosen uniformly among the family of reference elements. More precisely, we need the family $\{\hat{K} \,:\, K \in \mathcal{T}^n\}$ to form a compact subset

[4]The vector proxy associated with pullback of forms.

of

$$\left\{ [v_0 \cdots v_n] \mid \forall\, v_0, \ldots, v_n \in \mathbf{R}^n \ : \ \det \left[v_1 - v_0 | \cdots | v_n - v_0 \right] \neq 0 \right\} ,$$

and the function $\hat{K} \mapsto C(\hat{K})$ be continuous. The family of meshes needs to be **shape-regular**, or it could degenerate.

2.4 Rates

We want to prove that, for all $\boldsymbol{\tau} \in \mathbf{H}^1(\Omega)$

$$\|\boldsymbol{\tau} - \Pi_{0,K} \mathcal{R}_h^\epsilon \mathcal{E}_h^\epsilon \boldsymbol{\tau}\|_{0,\Omega} \preceq h \|\nabla \boldsymbol{\tau}\|_{0,\Omega} .$$

Strategy.

1. Prove $\|\mathcal{R}_h^\epsilon \mathcal{E}_h^\epsilon \boldsymbol{\tau} - \Pi_{0,K} \mathcal{R}_h^\epsilon \mathcal{E}_h^\epsilon \boldsymbol{\tau}\|_{0,K} \preceq h \|\nabla \boldsymbol{\tau}\|_{0,K}$
2. Prove $\|\boldsymbol{\tau} - \mathcal{R}_h^\epsilon \mathcal{E}_h^\epsilon \boldsymbol{\tau}\|_{0,K} \preceq h \|\nabla \boldsymbol{\tau}\|_{0,K}$
3. Conclude by using triangle inequality and summing over $K \in \mathcal{T}$.

The standard estimate for Raviart-Thomas on a reference element gives

$$\|\mathcal{R}^\epsilon \mathcal{E}^\epsilon \hat{\boldsymbol{\tau}} - \Pi_{0,\hat{K}} \mathcal{R}^\epsilon \mathcal{E}^\epsilon \hat{\boldsymbol{\tau}}\|_{0,\hat{K}} \leq \|\nabla \mathcal{R}^\epsilon \mathcal{E}^\epsilon \hat{\boldsymbol{\tau}}\|_{0,\hat{K}} .$$

It is simple to prove a Young's inequality:

Proposition 5.2.3. *For $\delta > 0$, a subset U of $\mathbb{R}^n$, and $u \in \mathrm{W}^{\ell,q}\Lambda^k(\mathcal{V}^\delta(U))$, there holds*

$$\|\nabla^\ell \mathcal{R}^\delta u\|_{\mathrm{L}^q(U)} \leq \|\nabla^\ell u\|_{\mathrm{L}^q(\mathcal{V}^\delta(U))} .$$

From where

$$\|\mathcal{R}^\epsilon \mathcal{E}^\epsilon \hat{\boldsymbol{\tau}} - \Pi_{0,\hat{K}} \mathcal{R}^\epsilon \mathcal{E}^\epsilon \hat{\boldsymbol{\tau}}\|_{0,\hat{K}} \leq \|\nabla \mathcal{E}^\epsilon \hat{\boldsymbol{\tau}}\|_{0,\mathcal{V}^\epsilon(\hat{K})} .$$

Scaling back to K we gain a power of h

$$\|\mathcal{R}_h^\epsilon \mathcal{E}_h^\epsilon \boldsymbol{\tau} - \Pi_{0,K} \mathcal{R}_h^\epsilon \mathcal{E}_h^\epsilon \boldsymbol{\tau}\|_{0,K} \preceq h \|\nabla \mathcal{E}_h^\epsilon \boldsymbol{\tau}\|_{0,\mathcal{V}_h^\epsilon(K)} .$$

Of course we cannot expect this to work for any extension $\mathcal{E}^\epsilon$. We shall require that $\mathcal{E}$ **preserves constants**[5], commutes with div and is uniformly stable

$$|\mathcal{E}_h^\epsilon \boldsymbol{\tau}|_{\ell,\mathcal{V}_h^\epsilon(K)} \preceq |\boldsymbol{\tau}|_{\ell,\mathcal{V}_h^\epsilon(K)\cap\Omega} , \quad \text{for} \quad 0 \leq \ell \leq 1.$$

This yields two types of estimates: for interior cells $K \in \mathcal{T}_i^n$ we get

$$\|\mathcal{R}_h^\epsilon \mathcal{E}_h^\epsilon \boldsymbol{\tau} - \Pi_{0,K} \mathcal{R}_h^\epsilon \mathcal{E}_h^\epsilon \boldsymbol{\tau}\|_{0,K} \preceq h \|\nabla \boldsymbol{\tau}\|_{0,\mathcal{V}_h^\epsilon(K)} ,$$

[5] In the general setting, we need to preserve polynomials up to the highest degree contained in the discrete space.

while for cells touching the boundary $K \in \mathcal{T}^n_\partial$

$$\|\mathcal{R}^\epsilon_h \mathcal{E}^\epsilon_h \boldsymbol{\tau} - \Pi_{0,K} \mathcal{R}^\epsilon_h \mathcal{E}^\epsilon_h \boldsymbol{\tau}\|_{0,K} \preceq h \|\nabla \boldsymbol{\tau}\|_{0,\mathcal{V}^\epsilon_h(K) \cap \Omega}\,.$$

Under quasi-uniformity of $\mathcal{T}$, we can choose ϵ so that $\mathcal{V}^\epsilon_h(K) \subset \mathcal{M}(K)$, so the overlap between the neighbourhoods $\mathcal{V}^\epsilon_h(K)$ is bounded independently of ϵ and h; summing over K we obtain

$$\|\mathcal{R}^\epsilon_h \mathcal{E}^\epsilon_h \boldsymbol{\tau} - \Pi_{0,K} \mathcal{R}^\epsilon_h \mathcal{E}^\epsilon_h \boldsymbol{\tau}\|_{0,\Omega} \preceq h \|\nabla \boldsymbol{\tau}\|_{0,\Omega}\,.$$

Now step 2. As usual, polynomial reproduction is key if we want to obtain approximation properties. In [3] Snorre noted that a simple requirement on the mollifier ρ is enough to guarantee that $\mathcal{R}^\epsilon$ satisfies this: if

$$\forall\, g \in P_r(\mathbf{R}^n) \quad \int_{\mathbf{R}^n} \rho\, g = g(0),$$

then we can prove

$$\forall\, \boldsymbol{\tau} \in \mathbf{P}_r(\mathbf{R}^n) \quad \mathcal{R}^\epsilon \boldsymbol{\tau} = \boldsymbol{\tau}.$$

Our case is $r = 0$, where this holds trivially because the mollifier integrates to one.

Let $\mathbf{T}^1 \boldsymbol{\tau}$ be the averaged Taylor polynomial of first order. Since it is a constant, and $\mathcal{R}^\epsilon_h \mathcal{E}^\epsilon_h$ preserves them, we have[6]

$$\begin{aligned}\|\boldsymbol{\tau} - \mathcal{R}^\epsilon_h \mathcal{E}^\epsilon_h \boldsymbol{\tau}\|_{0,K} &\leq \|\boldsymbol{\tau} - \mathbf{T}^1 \boldsymbol{\tau}\|_{0,K} + \|\mathcal{R}^\epsilon_h \mathcal{E}^\epsilon_h \mathbf{T}^1 \boldsymbol{\tau} - \mathcal{R}^\epsilon_h \mathcal{E}^\epsilon_h \boldsymbol{\tau}\|_{0,K} \\ &\preceq \|\boldsymbol{\tau} - \mathbf{T}^1 \boldsymbol{\tau}\|_{0,\mathcal{V}^\epsilon_h(K)} \preceq h \|\nabla \boldsymbol{\tau}\|_{0,\mathcal{M}(K)}\,,\end{aligned}$$

where for the last estimate we used Brenner and Scott's version of Bramble-Hilbert. The desired estimate then follows by summing over K.

We have arrived to the long desired estimate

$$\|\boldsymbol{\tau} - \Pi_{0,K} \mathcal{R}^\epsilon_h \mathcal{E}^\epsilon_h \boldsymbol{\tau}\|_{0,\Omega} \preceq h \|\nabla \boldsymbol{\tau}\|_{0,\Omega}\,.$$

This suggests defining $\mathbf{Q}^\epsilon_h := \Pi_{0,K} \mathcal{R}^\epsilon_h \mathcal{E}^\epsilon_h$ as the smoothed projection.

2.5 Problems?

We have constructed an operator $\mathbf{Q}^\epsilon_h$ that satisfies

$$\|\boldsymbol{\tau} - \mathbf{Q}^\epsilon_h \boldsymbol{\tau}\|_{0,\Omega} \preceq h \|\nabla \boldsymbol{\tau}\|_{0,\Omega}\,.$$

Further, we have taken care so that every involved operator commutes with div. Therefore[7]

$$\mathsf{div}\, \mathbf{Q}^\epsilon_h \boldsymbol{\tau} = \mathrm{Q}^\epsilon_h\, \mathsf{div}\, \boldsymbol{\tau}\,.$$

[6] Here we also should've obtained separate estimates for interior and boundary cells, but we have to finish on time!

[7] For simplicity, we assume a commuting projection Q^ϵ_h onto $\mathsf{DG}_0(\mathcal{T})$ was constructed. One usually does the construction for the entire complex simultaneously.

The problem is that $\mathbf{Q}_h^\epsilon$ is only guaranteed to preserve constants! Said differently, $\mathbf{Q}_h^\epsilon$ **is not a projection onto** $\mathbf{RT}_0(\mathcal{T})$.

There is a clever way of fixing this, that I also learned from Snorre, and it blew my mind...

The idea is to prove that $\mathbf{Q}_h^\epsilon|_{\mathbf{RT}(\mathcal{T})}$ is an isomorphism and then define

$$\Pi_h = (\mathbf{Q}_h^\epsilon|_{\mathbf{RT}(\mathcal{T})})^{-1}\mathbf{Q}_h^\epsilon\,.$$

The key point is that this preserves all the above properties! Namely, it commutes with div, is uniformly stable $\mathbf{L}^2 \to \mathbf{L}^2$ and has optimal rates of approximation. Our construction here is a bit different, so we still need to check invertibility.

We shall need two properties of $\mathbf{Q}_h^\epsilon$. First, it is bounded in $\mathbf{L}^2(\Omega)$ uniformly in h. Second, on discrete fields the smoothing error is uniformly (in h) small:

$$\forall\,\epsilon' > 0 : \exists \bar\epsilon : \forall\,\epsilon \le \bar\epsilon \;:\; \forall\,\boldsymbol{\tau} \in \mathbf{RT}_0(\mathcal{T}) \;:\; \|\boldsymbol{\tau} - \mathbf{Q}_h^\epsilon\boldsymbol{\tau}\|_{0,K} \le \epsilon'\|\boldsymbol{\tau}\|_{0,\mathcal{M}(K)}\,,$$

in fact with $\epsilon' \simeq \epsilon$. The rationale behind is that, for $\boldsymbol{\tau} \in \mathbf{RT}_0(\mathcal{T})$, the error

$$\boldsymbol{\tau} - \mathbf{Q}_h^\epsilon\boldsymbol{\tau} = \Pi_{0,\mathcal{T}}(\boldsymbol{\tau} - \mathcal{R}_h^\epsilon\mathcal{E}_h^\epsilon\boldsymbol{\tau})$$

only depends on the face moments $\int_f(\boldsymbol{\tau} - \mathcal{R}_h^\epsilon\mathcal{E}_h^\epsilon\boldsymbol{\tau})\cdot\mathbf{n}$, and $\boldsymbol{\tau}\cdot\mathbf{n}$ is continuous across f even though $\boldsymbol{\tau}$ is not.

Summing over K yields

$$\|\boldsymbol{\tau} - \mathbf{Q}_h^\epsilon\boldsymbol{\tau}\|_{0,\Omega} \le \sqrt{N}\,\epsilon'\|\boldsymbol{\tau}\|_{0,\Omega},$$

for all $\boldsymbol{\tau} \in \mathbf{RT}_0(\mathcal{T})$, equivalently

$$\|(I - \mathbf{Q}_h^\epsilon)|_{\mathbf{RT}(\mathcal{T})}\|_{\mathbf{L}^2\to\mathbf{L}^2} \le \sqrt{N}\,\epsilon'\,.$$

We fixed ϵ' freely, so we may put $\epsilon' = 1/(2\sqrt{N})$, which gives that $\mathbf{Q}_h^\epsilon|_{\mathbf{RT}(\mathcal{T})}$ is invertible and

$$\|(\mathbf{Q}_h^\epsilon|_{\mathbf{RT}(\mathcal{T})})^{-1}\| \le \frac{1}{1-1/2} \le 2\,.$$

Finally (!), we define

$$\Pi_h = (\mathbf{Q}_h^\epsilon|_{\mathbf{RT}(\mathcal{T})})^{-1}\mathbf{Q}_h^\epsilon : \mathbf{L}^1(\Omega) \to \mathbf{RT}_0(\mathcal{T})\,.$$

It is **uniformly bounded** $\mathbf{L}^2(\Omega) \to \mathbf{L}^2(\Omega)$ and **commutes with the divergence.**

We finish by verifying the rates. For $\boldsymbol{\tau} \in \mathbf{H}^1(\Omega)$, we have

$$\|\boldsymbol{\tau} - \Pi_h\boldsymbol{\tau}\|_{0,\Omega} \le \|\boldsymbol{\tau} - \mathbf{Q}_h^\epsilon\boldsymbol{\tau}\|_{0,\Omega} + \|\Pi_h\boldsymbol{\tau} - \mathbf{Q}_h^\epsilon\boldsymbol{\tau}\|_{0,\Omega}\,.$$

The first term is fine, and the second can be estimated as

$$\|\Pi_h\boldsymbol{\tau} - \mathbf{Q}_h^\epsilon\boldsymbol{\tau}\|_{0,\Omega} \preceq \|\mathbf{Q}_h^\epsilon|_{\mathbf{RT}(\mathcal{T})}(\Pi_h\boldsymbol{\tau} - \mathbf{Q}_h^\epsilon\boldsymbol{\tau})\|_{0,\Omega} = \|(\mathbf{Q}_h^\epsilon\boldsymbol{\tau} - \mathbf{Q}_h^\epsilon\mathbf{Q}_h^\epsilon\boldsymbol{\tau})\|_{0,\Omega} = \|\mathbf{Q}_h^\epsilon(I - \mathbf{Q}_h^\epsilon)\boldsymbol{\tau}\|_{0,\Omega}\,.$$

Then

$$\|\boldsymbol{\tau} - \Pi_h\boldsymbol{\tau}\|_{0,\Omega} \preceq \|\boldsymbol{\tau} - \mathbf{Q}_h^\epsilon\boldsymbol{\tau}\|_{0,\Omega} \preceq h\|\nabla\boldsymbol{\tau}\|_{0,\Omega}\,.$$

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